

4.1

$$a. SST = \sum (y_i - \bar{y})^2 = \sum y_i^2 - N\bar{y}^2 = 7825,34 - 20 \cdot (19,21)^2 = 445,458$$

$$R^2 = \frac{SSR}{SST} = \frac{275,47}{445,458} = 84,29\%$$

$$b. \hat{\sigma}^2 = \frac{\sum (\hat{e}_i)^2}{N-2} = \frac{SSE}{N-2} = \frac{(1-R^2) \cdot SST}{N-2} = \frac{(1-0,7911) \cdot 725,34}{18} = 8,4249$$

$$c. \sum (y_i - \bar{y})^2 = SST = 631,63$$

$$\sum \hat{e}_i^2 = SSE = 182,85$$

$$R = 1 - \frac{SSE}{SST} = 1 - \frac{182,85}{631,63} = 0,7105 \approx 71,05\%$$

4.3

$$a. \hat{y}_0 = 1,2 + 0,8x_i = 1,2 + 0,8 \cdot 4 = 4,4$$

$$b. \bar{x} = \frac{3+2+1-1+0}{5} = 1$$

$$\hat{\sigma}^2 = \frac{\sum (\hat{e}_i)^2}{N-2} = \frac{3,6}{3} = 1,2$$

$$\widehat{\text{var}}(f) = 1,2 \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right] = 2,52$$

$$\Rightarrow se(f) = \sqrt{2,52} \approx 1,5875$$

$$c. t_{0,975,3} = 3,182$$

Prediction interval (PI):

$$\hat{y}_0 - t_c se(f) \leq y_0 \leq \hat{y}_0 + t_c se(f)$$

$$4,4 - 3,182 \cdot 1,5875 \leq y_0 \leq 4,4 + 3,182 \cdot 1,5875$$

$$-0,6514 \leq y_0 \leq 9,4514$$

$$\Rightarrow 55\% \text{ PI for } y \text{ given } x_0 = 4 \text{ is } [-0,6514; 9,4514]$$

$$d. t_{0,995,3} = 5,841$$

PI:

$$4,4 - 5,841 \cdot 1,5875 \leq y_0 \leq 4,4 + 5,841 \cdot 1,5875$$

$$-4,87 \leq y_0 \leq 13,67$$

$$\Rightarrow 99\% \text{ PI for } y \text{ given } x_0 = 4 \text{ is } [-4,87; 13,67]$$

$$e. x = \bar{x} = 1 \rightarrow \hat{y}_0 = 1,2 + 0,8 \cdot 1 = 2$$

$$se(f) = \sqrt{1,2 \left(1 + \frac{1}{5} + 0 \right)} = 1,2$$

$$t_{0,975,3} = 3,182$$

$$PI : \hat{y}_0 - t_c se(f) \leq y_0 \leq \hat{y}_0 + t_c se(f)$$

$$2 - 3,182 \cdot 1,2 \leq y_0 \leq 2 + 3,182 \cdot 1,2$$

$$-1,82 \leq y_0 \leq 5,82$$

$\Rightarrow 95\%$ PI for y given $x = \bar{x}$ is $[-1,82 ; 5,82]$

$\Rightarrow$ PI (e) narrow than PI (c)

because x near $\bar{x} \rightarrow se(f)_{\min} \rightarrow$ PI narrow.